TrendPilot

A Local Agentic AI Assistant Using Ollama

AIRI Team PITB - AI Internship Task 2

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Repository	github.com/immadikram/agentic_ai_task_2

1. Project Overview

TrendPilot is a local agentic AI assistant built with Python and Ollama using the Gemma 3:4B model. Given a topic, platform, tone and output type, it plans its work, calls eight tools, reviews its own output, rewrites the caption if needed, and saves the final result as a Markdown file. It solves the problem of turning a raw project idea into a platform-ready content plan (idea, hook, caption, hashtags, reel script, title and thumbnail text) without any cloud API.

2. Tools and Technologies

Python	**Ollama**	**Gemma 3:4B**	**requests**
JSON memory	**Markdown file saving**	**FastAPI**	**Streamlit**
CLI	**ReportLab**	**GitHub**	

3. Agent Workflow

1	User input (topic, platform, tone, output type)
2	Agent builds a plan
3	Trend Idea Generator creates 5 angles
4	Agent picks the best angle and audience
5	Caption Writer
6	Hashtag Generator
7	Reel Script Generator
8	Title / Thumbnail Generator
9	Content Reviewer
10	If verdict is IMPROVE or score < 7: Caption Writer rewrites using the reviewer's feedback
11	File Saver
12	Memory Tool
13	Final output

Every tool call is printed as **[Tool Called]: name**.

4. Tools Implemented

Tool	What it does
Trend Idea Generator	Produces 5 specific content angles; uses memory to avoid repeating old angles.
Caption Writer	Writes the hook and caption following platform rules; also used for the improvement rewrite.
Hashtag Generator	Creates relevant hashtags, de-duplicated and limited per platform.
Reel Script Generator	Writes a 4-scene, ~30 second script with visuals and voiceover.
Title & Thumbnail Generator	Creates a catchy title and short thumbnail text.
Content Reviewer	Scores the content 1-10 and checks hook, clarity, length, tone, hashtags.
File Saver	Saves the final result to .md (or .json) in outputs/.
Memory Tool	Stores previous topics, platforms, tones and saved files in memory.json.

5. Demo and Outputs

Demos: CLI (**python demo.py**), Streamlit (**streamlit run app/streamlit_app.py**) and FastAPI (**uvicorn app.main:app**, **POST /generate-content**). Generated files are in **outputs/**.

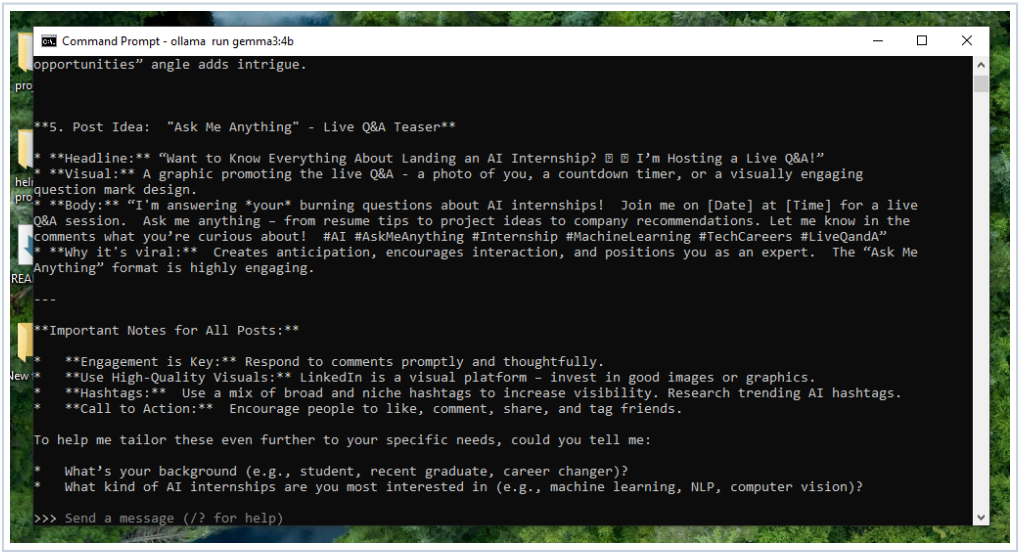


Figure 1: Ollama running the local model

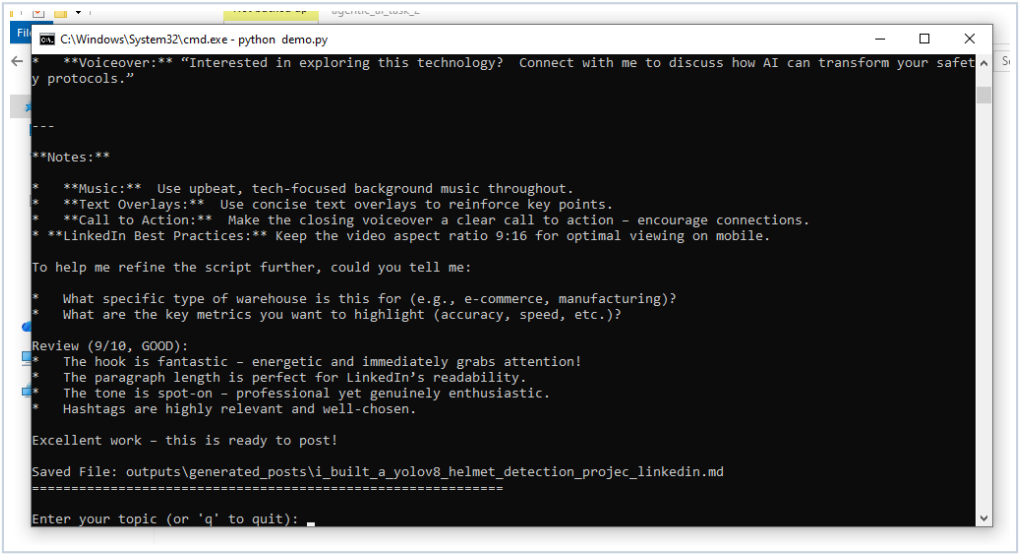


Figure 2: Agent output in the demo

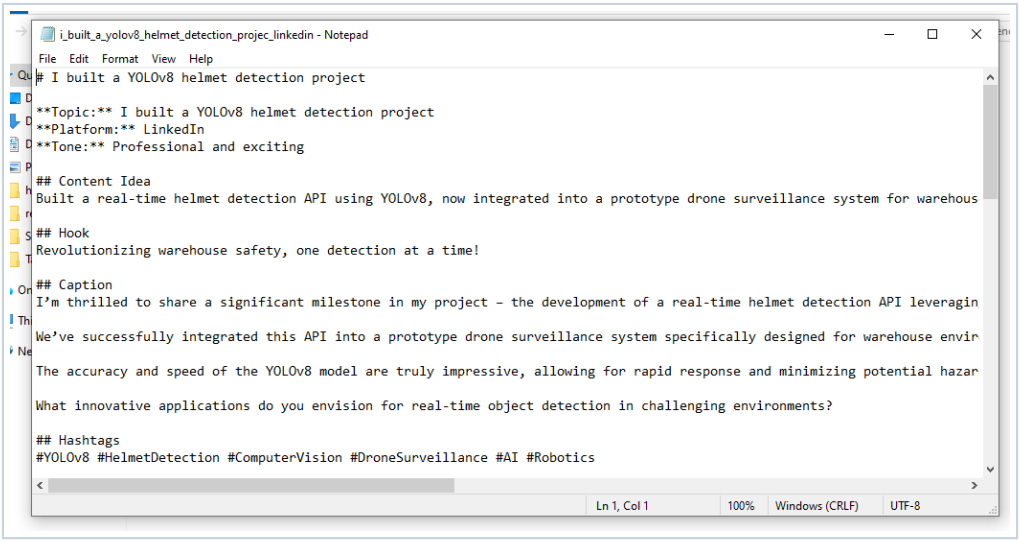


Figure 3: Saved Markdown output file

6. Testing (10 test cases)

Summary: I tested the agent on 10 topics across LinkedIn, Instagram, X/Twitter and YouTube Shorts, with tones from professional to meme style. Nine of the 10 outputs scored 8-9/10 from the reviewer. The weakest was test 6 (Instagram, meme style), which scored 6/10, so humour and meme prompts needed the most work.

#	Topic	Platform	Tone	Tools used	Score	Saved file	Useful?
1	YOLOv8 helmet detection	LinkedIn	Professional	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8/10	outputs/generated_posts/ **yolov8_helmet_detection_linkedin.md**	**Yes**
2	AI internship experience	LinkedIn	Motivational	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8/10	outputs/generated_posts/ **ai_internship_experience_linkedin.md**	**Yes**
3	Python learning journey	Instagram	Funny	Trend Idea Generator, Caption Writer, Reel Script Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8/10	outputs/generated_scripts/ **python_learning_journey_instagram.md**	**Yes**
4	Machine learning project	X/Twitter	Short and catchy	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8/10	outputs/generated_posts/ **machine_learning_project_x_twitter.md**	**Yes**
5	Data annotation struggles	LinkedIn	Humorous	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8/10	outputs/generated_posts/ **data_annotation_struggles_linkedin.md**	**Yes**
6	Model failed in production	Instagram	Meme style	Trend Idea Generator, Caption Writer, Reel Script Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	6/10	outputs/generated_scripts/ **model_failed_in_production_instagram.md**	**No***
7	Final year project	LinkedIn	Professional	Trend Idea Generator, Caption Writer, Hashtag Generator, Reel Script Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8/10	outputs/generated_posts/ **final_year_project_linkedin.md**	**Yes**
8	Computer vision project	YouTube Shorts	Exciting	Trend Idea Generator, Caption Writer, Reel Script Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8/10	outputs/generated_scripts/ **computer_vision_project_youtube_shor.md**	**Yes**
9	Debugging FastAPI	X/Twitter	Funny	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	8/10	outputs/generated_posts/ **debugging_fastapi_x_twitter.md**	**Yes**
10	Open-source contribution	LinkedIn	Inspirational	Trend Idea Generator, Caption Writer, Hashtag Generator, Title & Thumbnail Generator, Content Reviewer, File Saver, Memory Tool	9/10	outputs/generated_posts/ **open_source_contribution_linkedin.md**	**Yes**

*** Test 6:** Meme style was requested, but the reviewer scored the output 6/10 and asked for a rewrite.

> **Reviewer scores:** 9 of the 10 tests scored 8/10 or higher; the lowest was test 6 (6/10).

7. Error Analysis

Test case	Problem found	Possible reason	Fix applied / planned
Machine learning project (X/Twitter, Short and catchy)	Output was generic on the first draft. Reviewer flagged it (score 8/10): • Hook could be tighter - “AI Kicks!” is punchier. • “Grail kicks” is a bit verbose; simplify. • Consider a stronger call to action.	Prompt did not ask for specific details.	Asked for specific details in angles and relied on the reviewer rewrite loop.
Debugging FastAPI (X/Twitter, Funny)	Humor / meme style not convincing: humorous tone requested but reviewer scored it 8/10 and asked for a rewrite.	Model did not understand meme style from a tone word alone.	Added a humor instruction with an example to the caption and reel prompts.
Model failed in production (Instagram, Meme style)	Lowest score of all 10 tests (6/10); the reviewer asked for a rewrite and the result was marked not useful.	Likely the same cause as above: the model does not get meme style from a tone word alone.	Planned: add a meme example (setup + twist) to the reel prompt and re-test.
YOLOv8 helmet detection (LinkedIn, Professional)	Output invented details the user never gave: the input was only “I built a YOLOv8 helmet detection project”, but the saved post describes a drone surveillance system for warehouses (Figure 3).	Prompt did not tell the model to use only facts from the user’s input.	Planned: add “use only the facts given; use placeholders like [add accuracy %] for unknown details”.
YOLOv8 helmet detection (LinkedIn, Professional)	Reviewer gave 9/10 and “ready to post” to the output above, so it missed the invented details (Figure 2).	Reviewer judged tone and style only; it never compared the output with the original topic.	Planned: add a reviewer check “does the output contain claims not in the topic?”.
YOLOv8 helmet detection (CLI demo)	Output ended with chatty follow-up questions (“could you tell me what type of warehouse…”) and extra notes instead of only the final content (Figure 2).	Model answered like a chatbot; prompt did not say to return only the content.	Planned: add “return only the requested content, no questions or notes” to the prompts.
Computer vision project / YOLOv8 helmet (saved files)	Saved filenames were cut off mid-word, e.g. computer_vision_project_youtube_shor.md and i_built_a_yolov8_helmet_detection_projec_linkedin.md.	File name was truncated at a fixed character limit.	Planned: raise the limit or cut at a word boundary.

Suggested improvements

1. Add few-shot examples to prompts (especially humour / meme style).
2. Add validation checks (word count, character limit for X, hashtag count).
3. Add a user-feedback loop and a web-search tool.
4. Add a reviewer check for claims that are not in the user’s input (hallucination check).
5. Try a larger local model.

8. What I Learned

- How to run local LLMs with Ollama
- Difference between a chatbot and an agent
- Connecting tools to an LLM
- Prompt design for structured output
- Saving results and basic memory
- Testing and improving an agentic workflow

9. Future Improvements

- Web search
- Multi-agent workflow (planner / writer / reviewer)
- Voice input
- PDF export
- SQLite memory
- User feedback loop